
META-DIFFUSION: PLAN-ALIGNED RESULTS AND FITZPATRICK17K SOURCE SPECIFICITY

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ABSTRACT

We report the completion of a pre-registered experiment sequence designed to distinguish reusable population transport from genuinely source-task-specific transport. The sequence audits the data and frozen method (P0), raises Fitzpatrick17k task resolution from diseases to pathology families (P1), introduces hierarchical wrong-source controls (P2), tests a source-only task-separability explanation (P3), repeats the family protocol under a stronger Fitzpatrick I–III to V–VI shift (P4), audits the completed CIFAR package (P5), and applies a strict gate to downstream demographic augmentation (P6). On Fitzpatrick17k, correct-source transport improves energy MMD over a population-only control, but is statistically indistinguishable from wrong-task transport under both phenotype definitions. The frozen family classifier also fails its pre-specified per-family validity rule, so semantic nulls are classified as inconclusive rather than equivalence. Source-only task separability does not provide a supported explanation. In contrast, audited CIFAR evidence retains a clear correct-versus-wrong-source advantage at $K_T = 0$, survives FiLM conditioning, and supports an approximately linear task-space relation. The resulting claim is therefore deliberately narrow: source-specific coordinate transport is supported in the controlled CIFAR setting, whereas the frozen method learns primarily a shared phenotype offset on Fitzpatrick17k.

1 SCOPE AND REPORTING CONVENTION

The core model, training recipe, linear transport, and no-refinement headline inference were frozen before the new family-level outcomes were inspected. No new head, larger coordinate, nonlinear transport, or architecture rescue was introduced. All within-task comparisons use the same support draw, real target query, and DDIM noise. Repeats reduce Monte Carlo noise but are not treated as independent semantic tasks.

Raw-performance tables report absolute means $\pm$ 95% confidence-interval half-widths. A separate comparison table reports absolute paired differences. For lower-is-better distances, the comparison is

$$\Delta_d = d(\text{comparator}, \mathcal{D}_T) - d(\text{correct source+transport}, \mathcal{D}_T), \quad (1)$$

so positive values favour correct-source transport. For higher-is-better semantic or phenotype consistency, the subtraction is reversed, again making positive values favour correct transport. A superscript * is attached directly to a paired result only when its 95% interval excludes zero. No star is placed on an unpaired raw mean.

The reported measurements are: phenotype target share and semantic consistency (higher is better), and fixed phototype-statistic distance, sliced Wasserstein distance (SW), and energy MMD (lower is better). A frozen predictor is a measurement instrument, not ground truth. Failure of its independent validity gate prevents a semantic claim and is not replaced post hoc by a distributional metric.

2 P0: PREREQUISITE AND REPRODUCIBILITY GATE

The experiment used the verified SFU Fitzpatrick17k-Renamed archive as a byte source. Images were matched to the official CSV using the supplied MD5 hashes and filtered with the

Fitzpatrick17k-C allow-list. Known near-duplicate components were collapsed before task allocation. The official schema contains no subject identifier; consequently patient-level leakage cannot be checked directly, and hash/near-duplicate filtering is the strongest available safeguard.

Table 1: P0 dataset, environment, and frozen-protocol audit.

Audit item	Verified value
Official CSV / archive rows	16,577 / 16,577
Byte-verified rows retained by Fitzpatrick17k-C	11,394
Retained rows with Fitzpatrick I–VI	11,098
P1 eligible pathology families (target IV–VI)	6
P4 eligible pathology families (target V–VI)	5
P1 task-level folds / P4 cyclic folds	3 / 5
Training / evaluation seeds	12,345 / 4,321
Python / PyTorch / CUDA	3.12.13 / 2.11.0 / 12.8
GPU	NVIDIA RTX 5080, 16,303 MiB
Frozen coordinate dimension / source budget	$k = 16 / M_S = 32$

All manifests passed on-disk, population-side, taxonomy, duplicate, and capacity checks. Target support is disjoint from target query and nested across $K_T \in \{1, 2, 5, 10, 20\}$. The family-level semantic-instrument check remains an outcome-validity gate and was therefore not used to select tasks or folds.

P0 conclusion. The prerequisite gate passed. Data provenance, exclusions, task folds, frozen checkpoints, seeds, comparison direction, and independent statistical units were fixed before the new outcomes were inspected.

3 P1: PATHOLOGY-FAMILY SEMANTIC RESOLUTION

The task is the official `nine_partition_label`; source is Fitzpatrick I–III and target is IV–VI. Six families meet the pre-model requirement of at least 64 clean source and 45 clean target images. Three task-level folds test every family exactly once. Model selection uses only the validation family in each fold and chooses training steps 16,000, 10,000, and 10,000. Evaluation uses 96 generated images per cell, DDIM-50, eight paired repeats, and no target-time refinement.

3.1 ABSOLUTE ZERO-SHOT PERFORMANCE

Table 2: P1 raw performance at $K_T = 0$ over six outer-test pathology families. Entries are absolute family means $\pm$ 95% CI. Phenotype share and family consistency are higher-is-better; all three distances are lower-is-better.

Method	Phenotype share $\uparrow$	Family consistency $\uparrow$	Statistic dist. $\downarrow$	SW $\downarrow$	Energy MMD $\downarrow$
Source reuse	0.0000 $\pm$ 0.0000	0.1651 $\pm$ 0.3227	1.7452 $\pm$ 0.2089	0.3081 $\pm$ 0.0192	5.7946 $\pm$ 1.0078
Population only	0.0000 $\pm$ 0.0000	0.1641 $\pm$ 0.3216	3.0504 $\pm$ 0.2209	0.2683 $\pm$ 0.0110	6.8472 $\pm$ 1.0089
Wrong-family source + relation	0.0000 $\pm$ 0.0000	0.1656 $\pm$ 0.3230	1.7185 $\pm$ 0.2170	0.3064 $\pm$ 0.0200	5.7468 $\pm$ 1.0114
Correct-family source + transport	0.0002 $\pm$ 0.0004	0.1656 $\pm$ 0.3230	1.7188 $\pm$ 0.2178	0.3063 $\pm$ 0.0199	5.7436 $\pm$ 1.0026

Table 3: P1 paired effects at $K_T = 0$. Every entry is oriented so positive values favour correct-family transport. A star means the paired 95% CI excludes zero.

Comparator ($\Delta > 0$: correct-family transport)	Phenotype share	Family consistency	Statistic dist.	SW	Energy MMD
Source reuse	+0.00022 $\pm$ 0.00043	+0.00043 $\pm$ 0.00054	+0.02635 $\pm$ 0.01482*	+0.00178 $\pm$ 0.00218	+0.05099 $\pm$ 0.05141
Population only	+0.00022 $\pm$ 0.00043	+0.00152 $\pm$ 0.00191	+1.33162 $\pm$ 0.08916*	−0.03795 $\pm$ 0.02043*	+1.10360 $\pm$ 0.49696*
Wrong-family source + relation	+0.00022 $\pm$ 0.00043	+0.00000 $\pm$ 0.00000	−0.00029 $\pm$ 0.00160	+0.00007 $\pm$ 0.00091	+0.00317 $\pm$ 0.01925

The semantic evaluator obtains overall fold accuracies of 67.6–68.7%, but its minimum outer-test-family accuracies are 0.0%, 0.0%, and 15.3%, below the locked 25% per-family threshold. Hence the primary correct-minus-wrong semantic effect of exactly zero is uninformative rather than evidence of equivalence. The independently valid secondary correct-versus-wrong effects all include zero.

3.2 TARGET-ONLY DATA-EFFICIENCY COMPARISON

Because the headline correct, wrong, source-reuse, and population-only coordinates receive no target-time refinement, they are invariant to K_T . Nonzero target budgets therefore test the target-only comparator rather than further adapting transport.

Table 4: P1 target-only raw distances and paired effects against the fixed correct-family transport. Within each metric, the first row is target-only raw performance and the second is $\Delta = d(\text{target only}) - d(\text{correct transport})$; positive supports correct transport.

Metric	Quantity	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Statistic dist.	Target only raw	1.7322 ± 0.2193	1.7306 ± 0.2201	1.7412 ± 0.2117	1.7369 ± 0.2147	1.7356 ± 0.2158
	Paired Δ (+ = correct)	$+0.0134 \pm 0.0227$	$+0.0118 \pm 0.0127$	$+0.0224 \pm 0.0119^*$	$+0.0180 \pm 0.0080^*$	$+0.0168 \pm 0.0092^*$
SW	Target only raw	0.3045 ± 0.0194	0.3055 ± 0.0191	0.3041 ± 0.0190	0.3047 ± 0.0204	0.3039 ± 0.0194
	Paired Δ (+ = correct)	-0.0018 ± 0.0037	-0.0007 ± 0.0012	$-0.0022 \pm 0.0018^*$	$-0.0016 \pm 0.0014^*$	$-0.0024 \pm 0.0020^*$
Energy MMD	Target only raw	5.7208 ± 0.9739	5.7303 ± 0.9850	5.6749 ± 0.9871	5.7091 ± 1.0350	5.6900 ± 1.0112
	Paired Δ (+ = correct)	-0.0228 ± 0.0745	-0.0133 ± 0.0315	$-0.0687 \pm 0.0379^*$	-0.0345 ± 0.0374	$-0.0536 \pm 0.0371^*$

The distance metrics disagree in direction, and target-only itself changes little with K_T . Consequently, no scientifically meaningful “transport at $K_T = 2$ equals target-only at $K_T = 10$ ” statement is supported.

P1 conclusion. Raising semantic resolution to pathology families does not recover a source-specific advantage. The primary result is inconclusive because its evaluator fails the validity gate; valid distributional diagnostics independently show no resolved correct-versus-wrong effect.

4 P2: HIERARCHICAL WRONG-SOURCE CONTROL

The frozen disease-level checkpoint is evaluated on four test diseases with correct disease, wrong disease in the same pathology family, wrong disease in a different family, source reuse, and population-only conditions. Wrong sources are deterministic and selected without using model outcomes.

Table 5: P2 raw zero-shot performance over four disease tasks. Consistency measures are higher-is-better; distances are lower-is-better.

Method	Phenotype share $\uparrow$	Disease consistency $\uparrow$	Statistic dist. $\downarrow$	SW $\downarrow$	Energy MMD $\downarrow$
Source reuse	0.1390 ± 0.0103	0.0199 ± 0.0224	2.5248 ± 0.4420	0.3642 ± 0.0448	7.1765 ± 1.8278
Population only	0.0000 ± 0.0000	0.0000 ± 0.0000	4.0465 ± 0.3111	0.2867 ± 0.0094	8.4246 ± 1.4825
Wrong disease, same family	0.1423 ± 0.0101	0.0199 ± 0.0224	2.5020 ± 0.4478	0.3638 ± 0.0454	7.1466 ± 1.8509
Wrong disease, different family	0.1426 ± 0.0098	0.0205 ± 0.0231	2.5025 ± 0.4462	0.3639 ± 0.0454	7.1444 ± 1.8488
Correct disease + transport	0.1419 ± 0.0094	0.0199 ± 0.0221	2.5030 ± 0.4480	0.3638 ± 0.0456	7.1429 ± 1.8566

Table 6: P2 hierarchical paired effects at $K_T = 0$. Positive values favour the first named condition. A star denotes a paired 95% interval excluding zero.

Comparison ($\Delta > 0$: first condition)	Phenotype share	Disease consistency	Statistic dist.	SW	Energy MMD
Correct vs. same-family wrong	-0.00033 ± 0.00283	$+0.00000 \pm 0.00104$	-0.00101 ± 0.00777	$+0.00009 \pm 0.00062$	$+0.00365 \pm 0.01937$
Same-family vs. different-family wrong	-0.00033 ± 0.00191	-0.00065 ± 0.00074	$+0.00053 \pm 0.00532$	$+0.00004 \pm 0.00038$	-0.00219 ± 0.01319
Correct vs. population only	$+0.14193 \pm 0.00938^*$	$+0.01986 \pm 0.02214$	$+1.54346 \pm 0.14526^*$	$-0.07706 \pm 0.03919^*$	$+1.28164 \pm 0.52335^*$

The disease instrument reaches 20.8% overall but only 53.1%, 0%, 6.2%, and 0% on the four individual test diseases. It is therefore unsuitable for a strong semantic ordering. The correct/same-family/different-family distance contrasts are also numerically small and all include zero. Population-only again produces low pixel SW while failing phenotype and energy MMD checks, demonstrating why no single pixel metric is sufficient.

P2 conclusion. Neither correct $>$ same-family wrong $>$ different-family wrong nor a family-only ordering is resolved. The null hierarchy is not an equivalence result.

5 P3: OUTCOME-INDEPENDENT SOURCE-TASK SEPARABILITY

Official ImageNet ResNet-18 penultimate features were extracted from source-population images only, L2-normalised per image, and averaged by disease. Separation is Euclidean distance to the pre-assigned different-family wrong source from P2.

Table 7: P3 source-only separation and held-out semantic effect.

Target disease	Separation	Correct-minus-wrong effect
Basal cell carcinoma	0.2488	0.0000
Neurofibromatosis	0.2807	0.0000
Pityriasis rosea	0.3289	−0.0026
Vitiligo	0.2971	0.0000

Spearman $\rho = -0.775$ with a task-bootstrap interval $[-1.000, -0.775]$. Only 6,793 of 10,000 bootstrap draws have defined rank variance because $n = 4$ and three outcome effects are exactly zero. Moreover, the outcome is measured by the weak disease instrument described above.

P3 conclusion. Greater source-task separability does not predict a larger correct-source advantage in the locked analysis. The result is unsupported/unresolved, not a negative causal mechanism, and task identifiability should not be the paper’s central explanation.

6 P4: STRONGER PHENOTYPE SEPARATION

P4 was triggered because P1 was inconclusive and the pre-outcome counts were adequate. The only domain change is target Fitzpatrick V–VI; source remains I–III. Five families are tested in five cyclic outer folds, with each family’s two appearances averaged before inference. Validation-only selection chooses steps 16,000, 4,000, 4,000, 4,000, and 10,000. The architecture is a 32×32 additive-basis small U-Net (0.72M parameters), set encoder (0.33M), linear transport (0.3K), coordinate dimension 16, and no target-time refinement.

Table 8: P4 raw performance at $K_T = 0$ over five pathology families. The stronger target is Fitzpatrick V–VI.

Method	Phenotype share $\uparrow$	Family consistency $\uparrow$	Statistic dist. $\downarrow$	SW $\downarrow$	Energy MMD $\downarrow$
Source reuse	0.0311 ± 0.0359	0.2033 ± 0.3807	1.8099 ± 0.2613	0.2649 ± 0.0446	4.5031 ± 0.7991
Population only	0.0000 ± 0.0000	0.1997 ± 0.3909	2.2577 ± 0.4548	0.2527 ± 0.0275	5.4122 ± 1.1998
Wrong-task source + relation	0.0320 ± 0.0369	0.2033 ± 0.3807	1.7943 ± 0.2765	0.2637 ± 0.0454	4.4233 ± 0.8218
Correct source + transport	0.0315 ± 0.0363	0.2031 ± 0.3808	1.7950 ± 0.2769	0.2638 ± 0.0461	4.4244 ± 0.8362

Table 9: P4 paired effects at $K_T = 0$. All entries are oriented so positive values favour correct-source transport.

Comparator ($\Delta > 0$: correct transport)	Phenotype share	Family consistency	Statistic dist.	SW	Energy MMD
Source reuse	$+0.00039 \pm 0.00051$	-0.00013 ± 0.00102	$+0.01494 \pm 0.01616$	$+0.00117 \pm 0.00272$	$+0.07874 \pm 0.08272$
Population only	$+0.03151 \pm 0.03635$	$+0.00339 \pm 0.01614$	$+0.46273 \pm 0.46930$	-0.01112 ± 0.02383	$+0.98779 \pm 0.70849^*$
Wrong-task source + relation	-0.00052 ± 0.00074	-0.00013 ± 0.00026	-0.00068 ± 0.00273	-0.00003 ± 0.00134	-0.00107 ± 0.03907

The phenotype instrument is useful on outer-test families (81.2–92.7%, chance 50%). The semantic instrument has 69.9–70.5% overall accuracy, yet worst-family accuracies of 0.0%, 0.0%, 0.0%, 12.7%, and 2.5%, so every fold fails the locked per-family rule. Pooled KID is consequently skipped. Correct transport significantly improves energy MMD over population-only, but the correct-versus-wrong-task effect is -0.0011 ± 0.0391 and is unresolved. The same near-zero pattern holds for every other valid secondary metric.

Table 10: P4 target-only raw distances and paired effects against fixed correct transport. For each metric, $\Delta = d(\text{target only}) - d(\text{correct transport})$; positive values support correct transport.

Metric	Quantity	$K_T = 1$	$K_T = 2$	$K_T = 5$	$K_T = 10$	$K_T = 20$
Statistic dist.	Target only raw	1.8082 ± 0.2694	1.8050 ± 0.2706	1.8044 ± 0.2691	1.7997 ± 0.2702	1.7972 ± 0.2689
	Paired Δ (+ = correct)	$+0.0132 \pm 0.0109^*$	$+0.0100 \pm 0.0121$	$+0.0094 \pm 0.0086^*$	$+0.0047 \pm 0.0071$	$+0.0022 \pm 0.0106$
SW	Target only raw	0.2597 ± 0.0413	0.2622 ± 0.0440	0.2624 ± 0.0443	0.2632 ± 0.0449	0.2630 ± 0.0447
	Paired Δ (+ = correct)	-0.0041 ± 0.0052	-0.0016 ± 0.0022	-0.0013 ± 0.0020	-0.0006 ± 0.0014	-0.0008 ± 0.0017
Energy MMD	Target only raw	4.3571 ± 0.7399	4.3970 ± 0.7715	4.4157 ± 0.7973	4.4171 ± 0.7974	4.3974 ± 0.7895
	Paired Δ (+ = correct)	-0.0673 ± 0.1207	-0.0274 ± 0.0679	-0.0086 ± 0.0460	-0.0073 ± 0.0406	-0.0270 ± 0.0576

P4 conclusion. A larger phenotype gap does not recover disease-family-specific source use. The model learns a reusable population shift relative to population-only on energy MMD, but obtains the same shift from the wrong source task. Under the registered stop rule, demographic model tuning ends here.

7 P5: FROZEN CIFAR PACKAGE

P5 was audited from existing raw per-episode artifacts rather than re-estimated opportunistically. The saved files contain matched episodes and paired intervals.

Table 11: Audited CIFAR evidence. All stated improvements are paired; starred rows have a 95% interval excluding zero in the underlying artifact.

Plan item	Verified result
$K_T = 0$ source specificity	Correct transport improves SW over wrong source by $0.0386 \pm 0.0141^*$ and fine-class consistency by $0.0597 \pm 0.0204^*$.
FiLM generality	Correct transport beats shuffled, mean-source, and relation-only controls on SW and semantic consistency at $K_T = 1, 5, 20$.
Linear relation	Linear transport is not worse than the nonlinear MLP and improves SW by 0.0093, 0.0112, and 0.0087 at $K_T = 1, 5, 20$.
Basis vs. FiLM	Transported generation quality is unresolved as different. Basis uses 55,344 conditioning parameters versus 118,784 for FiLM (46.6%).
Full fine-tuning	Transport is significantly safer at $K_T = 1, 5$; full fine-tuning becomes significantly better at $K_T = 20$.
25% training tasks	In the frozen one-seed run, basis improves SW by 0.0181–0.0203 and energy MMD by 0.4668–0.4975 over FiLM at $K_T = 1, 5, 20$.

Latency, FLOPs, and peak-memory numbers were not present as matched measurements and are not inferred from parameter count. The 25% task result is limited to one training seed.

P5 conclusion. The controlled core survives audit: source identity matters at zero target examples, the relation is approximately linear, its benefit concentrates in the low-data regime, and it generalises across conditioning mechanisms. The additive basis is an efficiency result, not a demonstrated generation-quality winner over FiLM.

8 P6: OPTIONAL DOWNSTREAM DEMOGRAPHIC AUGMENTATION

P6 was **not run—gate not met**. Its strict trigger required a clearly positive P1 correct-versus-wrong family-semantic result. P1 instead failed the semantic-instrument validity rule and showed no resolved correct-versus-wrong advantage on the valid secondary metrics. Launching or tuning a downstream classifier would therefore violate the plan.

9 DISCUSSION

The Fitzpatrick pattern is consistent across four complementary analyses. First, family resolution does not separate correct from wrong source coordinates. Second, wrong diseases within and outside a family are indistinguishable. Third, a frozen source-only separability score does not predict

the source advantage. Fourth, increasing the phenotype gap to V-VI improves neither task-source specificity nor semantic-instrument validity. Training diagnostics independently show that a shared mean coordinate captures nearly all coordinate gain: shuffled task coordinates and correct coordinates yield effectively identical losses.

This does not imply that demographic transport is impossible, nor does it prove that all source tasks are equivalent. The semantic evaluator fails on several held-out families and diseases, the number of independent tasks is small, and patient identifiers are unavailable. It does show that the frozen method and available measurements do not justify a real-data source-specific claim. In particular, a significant improvement over population-only is insufficient: wrong-task transport achieves the same improvement, identifying a shared population offset rather than a task-conditioned relation.

10 CONCLUSION

The executable plan is complete through its registered stopping conditions. On CIFAR, META-DIFFUSION supports the sequence

source identity matters $\rightarrow$ an approximately linear relation exists
 $\rightarrow$ benefit is largest under target scarcity $\rightarrow$ the principle survives FiLM.

On Fitzpatrick17k, the corresponding source-specific demographic extension is not supported. Correct and wrong task sources remain unresolved at disease and family resolution, including under the stronger V-VI target. The strongest defensible interpretation is therefore that the frozen model learns a shared phenotype shift but does not demonstrate disease-specific coordinate transport. The invalid semantic instrument requires the headline family result to remain inconclusive, not equivalent. No demographic architecture rescue or downstream augmentation is warranted under the locked protocol.

A FROZEN CONFIGURATION AND ARTIFACT MAP

Table 12: Frozen family-level training and evaluation configuration.

Component	Value
Image resolution / backbone width	$32 \times 32 / 32$
Conditioning / coordinate dimension	additive basis / 16
Transport	linear, approximately 0.3K parameters
Training	20,000 steps, AdamW, $\text{lr } 2 \times 10^{-4}$
EMA / training seed	0.999 / 12,345
Source support	$M_S = 32$
Target budgets	$K_T = 0, 1, 2, 5, 10, 20$
Generation	DDIM-50, $\eta = 0$, 96 images/cell
Paired repeats / evaluation seed	8 / 4,321
Headline target-time refinement	disabled

Machine-readable raw cells, family/disease-level values, checkpoint-selection records, manifests, hashes, commands, and complete tables are stored under `experiments/meta_diffusion_final/`. The principal files are `protocol.lock.md`, `prerequisite.audit.md`, `metrics/p1_family_aggregate.json`, `metrics/p2_hierarchical.json`, `metrics/p3_task_separability.json`, `metrics/p4_v.vi_aggregate.json`, and `runs/execution.log.md`. Debug runs are explicitly named and excluded from every reported estimate.